

Siyuan Feng

BSc Mathematics, University of Warwick | Machine Learning Theory

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RESEARCH INTERESTS

Machine learning theory; optimization and training dynamics; hyperparameter transfer; scaling laws; learning-rate schedules.

EDUCATION

University of Warwick

Sep 2024 – Jun 2027 (expected)

BSc Mathematics

- Year 1 average: 82.5%; Year 2 average: 82.0% (First-Class level; GPA equivalent: 3.94/4.00).

RESEARCH EXPERIENCE

Researcher, Machine Learning Theory

Jul 2026 – Present

Coauthored with Prof. Fanghui Liu | Manuscript in preparation

Warmup–Stable–Decay Schedule Transfer under μP Across Width and Training Steps

- Proved that, for any fixed number of training steps, final loss becomes stable across model widths for all choices of WSD’s three schedule parameters in a deep nonlinear μP network. This supports transfer of near-optimal schedules and final performance even when several schedules are equally good.
- Proved that results from only one training length are not enough in general: neither its best schedule nor complete information about final loss across all schedule choices can always determine the corresponding result at a different training length.
- Developed a method that uses two shorter training runs to predict final loss across all WSD schedule choices in a longer run. Proved a uniform error bound and showed how the prediction can be reused for wider models with explicit bounds on final performance.

Warwick Undergraduate Research Support Scheme (URSS) Researcher

May 2026 – Jul 2026

Supervised by Prof. Fanghui Liu | Completed manuscript

Two-Step Optimal Learning-Rate Geometry under μP

- Derived an exact low-dimensional description of the two-step training dynamics and an explicit infinite-width formula for how final loss depends on the learning rate in a deep linear μP network.
- Proved that the optimal learning rate is unique for almost all datasets with a positive-definite Gram matrix, while also constructing full-rank datasets with exactly three equally good optimal learning rates.
- Showed that the optimal set remains stable under small changes to the data and that the corresponding local minima remain visible at finite width; completed a full research manuscript.

AWARDS & RESEARCH FUNDING

£500 Research Bursary — University of Warwick Undergraduate Research Support Scheme

2026

Awarded in support of the project *Two-Step Optimal Learning-Rate Geometry under μP* .

£75 Prize — 2025 Quant Championship

2025

Placed 4th out of 200 participants.

RESEARCH METHODS & TECHNICAL SKILLS

- Theory** — Tensor Programs and infinite-width neural-network analysis; stability of final-loss functions and optimal solutions; spectral and polynomial methods; approximation error for finite-step training.
- Research computing** — **Python (NumPy, SciPy, PyTorch, pandas, Matplotlib) and MATLAB:** Built neural-network training pipelines, grid-based hyperparameter searches, simulations, finite-width checks, and reproducible figure-generation workflows.
- Research workflow** — **LaTeX, Git, Linux, Jupyter, and Conda:** Prepared mathematical manuscripts and maintained version-controlled, reproducible computational workflows.

INTERNSHIP, TEACHING & OUTREACH EXPERIENCE

Tianma Microelectronics Co., Ltd. – Algorithm Engineering Intern

Aug 2025 – Sep 2025

- Built a pixel-level E-paper video ghosting simulation pipeline in MATLAB and Python, covering frame extraction, drive mapping, and grayscale output.
- Replaced pixel-wise loops with NumPy vectorization in a history-frame-difference algorithm, reducing single-run simulation time by approximately 85%.

A-Level Mathematics & Cambridge STEP Tutor

Apr 2024 – Aug 2024

- Delivered 100+ hours of mathematics tutoring; supported students who later received offers from Oxford, Imperial College London, NUS, and other universities.

Calculus Education Content Creator, Bilibili

Ongoing

- Built and operate a calculus education channel with 12,000+ followers, producing structured video lessons on core concepts, mathematical derivations, and problem-solving techniques.